Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/10/2020
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Study the Molecular Mechanisms That Underlie Directional Root-Growth Responses to Low-Speed 2-Dimensional Clinorotation		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
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Task Description:	As key components of bioregenerative life-support systems, plants will be exposed to multiple stresses associated with spaceflight. Experiments carried out in Space have dramatically improved our understanding of plant responses to microgravity. In turn, this novel knowledge has allowed the design of better culturing methods that enable plant development and reproduction in Space. Yet, to design even better growth strategies and develop optimized genotypes favorable to plant growth and productivity during spaceflight, we need to better understand the molecular machinery that contribute to the perception of, and response to gravity and/or microgravity. Unfortunately, the large cost associated with Space experimentation and limited accessibility to the resources needed for such experimentation, limit our ability to acquire this new knowledge. Consequently, utilization of devices that mimic microgravity is viewed as a viable alternative. The 2D-clinostat has often been utilized for this purpose. By constantly changing the orientation of biological specimens relative to gravity, these devices prevent gravity-vector detection, thereby triggering behavioral responses that are similar to those expressed under true microgravity. The 2D-clinostat has often been used to estimate the sensitivity of plant organs to different doses of gravi-stimulation. In this case, low doses of gravi-stimulation are provided at the beginning of the experiment. Stimulated seedlings are subsequently clinorotated for 4-5 hours, allowing for a curvature to develop in response to the initial stimulus. Yet, the data derived from these analyses are likely to be significant only if the clinorotation does not introduce artifacts of its own. In experiments involving Arabidopsis thaliana seedlings, we previously observed a strong clinostat effect on root skewing on hard agar surfaces. Follow-up experiments involving Brachypodium distachyon confirmed a clear impact of low-speed clinorotation on directional root-tip bending. These experiments also suggested an important role for the first cycle of clinorotation in setting up a specific direction of gravistimulation as long as it is provided for a sufficiently long time period. The general objective of this research project will be to investigate the effect of low speed clinorotation on Brachypodium root growth. Specific objectives include: Aim 1: Develop high throughput time-lapse imaging to describe the development of B. distachyon root curvature responses to clinorotation at high spatio-temporal resolution. Aim 2: Use transgenic B. distachyon seedlings that express auxin-level, auxin-function, and auxin-efflux carrier reporters to investigate the effect of clinorotation on auxin distribution and signaling along the root tip, and on the distribution of auxin transporters between flanks of clinorotated root tips. Aim3: Use a combination of Genome Wide Association Study (GWAS) and RNA Seq strategies to identify candidate genes that may contribute to root-tip response to clinorotation, and use reverse genetics to investigate the role(s) of such candidates and of gravitropism-related genes in the process. Overall, this project will improve our understanding of how low-speed clinorotation affects root growth behavior. While our observations suggest a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system, they also suggest a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. This project will introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, while also putting a strong emphasis on undergraduate and K-12 education and public outreach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will improve our understanding of how low-speed clinorotation affects root growth behavior, while re-emphasizing a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system. Our research is likely to result in the development of a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. In the longer term, the basic information acquired during these studies should enable the design of plants that are better suited to function as key components of bioregenerative life support systems for long-term space exploration missions, either during travel under microgravity, or exposed to distinct gravity environments on exotic planets. They may also allow the development of novel crop cultivars with increased productivity in natural and/or agricultural ecosystems on Earth. With this project, we are able to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, in addition to developing effective educational materials to introduce K-12 and undergraduate students, as well as the general public, to the difficult concepts of gravity perception by plants and its importance for agricultural production on Earth and during spaceflight.
	When Arabidopsis thaliana or Brachypodium distachyon seedlings growing on the surface of agar-based media are positioned on a clinostat rotating along an axis that is perpendicular to the growth surface, their roots tend to curve in a direction that is dictated by the polarity of rotation, a behavior we previously named “clinostat effect.” Last year, we showed this effect occurs only when the plates are clinorotated at low speed. For wild type Columbia Arabidopsis seedlings, roots will display a strong clinostat effect at speeds of clinorotation equal to or below 1 rpm. We also showed that the direction of root curving due to the clinostat effect is dictated by the polarity of clinorotation. Interestingly, this polarity depends of the direction of gravistimulation imposed on the root tip during a phase of the rotation cycle that is long enough to trigger a response. Indeed, if 1-rpm clinorotation is initiated with wild type Col seedlings positioned vertically, their root tips will curve in a direction that is dictated by the gravistimulus provided during the first ½ turn of clinorotation (from vertical down to vertical up). However, if the clinorotation is initiated with seedlings positioned horizontally on the clinostat, the direction of curvature will not be dictated by the direction of gravistimulation during the initial 1/4 turn of rotation (15 seconds when the seedlings rotate from an horizontal to a vertical orientation), but by the gravistimulus provided during the next ½ turn phase (30 sec from vertical down to vertical up). This observation suggests that the clinostat effect is a consequence of gravistimulation during clinorotation, where the polarity of gravistimulation during the initial phase of rotation that provides a sufficiently long gravistimulus dictates the eventual direction of tip curvature, whereas subsequent rounds of rotation simply allow that curvature to proceed. If our hypothesis is correct, we expect mutations that alter root gravisensitivity will also affect the clinostat effect. Last year, we took advantage of several Arabidopsis thaliana mutations that affect root gravisensitivity to test this hypothesis. These included mutations in the PHOSPHOGLUCOMUTASE (PGM) gene, which encodes an enzyme necessary for starch synthesis and statolith sedimentation in the statocytes, mutations in TOC132, which encodes a protein that contributes to protein import into the plastids and interacts with ARG1 to orchestrate gravisensing, and mutations in the LAZY 2, 3, and 4 genes, which also contribute to early phases of gravity signal transduction in the statocytes, where they dictate response polarity. These preliminary experiments showed that these mutations affect the clinostat effect, in

Task Progress:	support of our model. This year, we modified our clinostat system to increase its throughput, and continued our comparative analysis of clinostat effect between wild type and mutant Arabidopsis seedlings. We also initiated a more careful quantification of the minimal time of gravistimulation during clinorotation needed to trigger the clinostat effect in Arabidopsis. These studies suggest a novel method to estimate root gravisensitivity based on the clinostat effect. Building a higher-throughput clinostat: To increase our ability to clinorotate multiple genotypes at the same time, we collaborated with Kate Baldwin (Freelance Scientific Designer) and Ben Vesperman (Engineer) to build a new clinostat that can rotate two plate-holding boxes along its axis (each box potentially carrying 6 square petri dishes). The boxes carrying the plates can be inserted on the clinostat axis in such a way that they are oriented with the plane perpendicular or parallel to the axis of rotation. This clinostat is driven by a step-motor that can be programmed to rotate at multiple speeds, thereby affording great experimental flexibility with our plant materials. This clinostat is inexpensive, allowing us to easily build multiple units. Such units will be used next year to evaluate the sensitivity of multiple Brachypodium accessions to the clinostat effect. Mathematical modeling of root curvature responses to clinorotation: The classical method to evaluate plant-organ gravi-sensitivity involves gravistimulating the plants for short time periods, then clinorotating them for a few hours to allow development of a curvature response to the initial stimulus. This leads to different levels of curvature depending on the initial period of stimulation. The data are then fit to a mathematical model that allows estimation of gravisensitivity. If a logarithmic model is chosen, extending the best-fit model to the X-axis allows estimation of the presentation time, which is defined as the minimal time of gravistimulation needed to allow the plant organ to initiate a productive curvature. If a hyperbolic model is used, the function will cross the X-axis at the origin. In this case, the slope of the curve at the origin can be used to estimate gravisensitivity. To further investigate the possible use of the clinostat effect to evaluate gravisensitivity, we initiated experiments aimed at evaluating the maximal speed of clinorotation needed for specific genotypes to develop a root tip curvature upon clinorotation, and compare these relative values to those estimated by classical presentation-time analyses. We first evaluated the presentation times of roots from wild type Col and toc132-2 mutant seedlings growing on 1% agar surfaces, using the classical method described above. Results indicated presentation times of 2.4 and 6.4 min for Col and toc132-2 mutant roots, respectively. Hence, toc132-2 roots are less sensitive to gravistimulation than wild type Col roots under our growth conditions. We next evaluated the effect of clinorotation speed on Arabidopsis root tip curvature (without initial gravistimulation), subjecting wild type and toc132-2 mutant seedlings to multiple clinorotation speeds ranging from 0.2 rpm to 2 rpm in separate experiments. The results indicated that wild type Col roots start curving in response to clinorotation at speeds of 1 rpm or below, with a maximal effect at 0.5 rpm. toc132 mutant roots, on the other hand, did not curve unless the clinostat rotated at a speed of 0.2 rpm. These results conclusively show that toc132 is less sensitive to the clinostat effect than wild type, and suggest it is also less gravi-sensitive than wild type, in agreement with our initial presentation-time studies. Hence, the clinostat may be a sensitive platform with which to investigate root sensitivity to gravistimulation. As an initial step to implement this novel platform, we adopted a mathematical modeling method similar to the one used in presentation-time analysis to quantify wild type Col root sensitivity to clinorotation, fitting the root tip curvature data obtained at various clinorotation speeds to a logarithmic model, and extending this model to its intersection with the time-axis. In this model, time is defined by half the periodicity of clinorotation. Indeed, when seedlings start clinorotation in a vertical orientation, their roots are gravistimulated in the same direction for ½ turn: 30 sec for 1 rpm; 1 min for 0.5 rpm; etc. The intersection of the model with the time axis should provide a valid evaluation of the minimal time (hence maximal speed) needed during a clinorotation cycle to allow fixation of root-tip curvature direction (as a clinostat-effect analogy to the presentation time parameter used in the classical model). Preliminary results obtained with wild type Col roots identified a clinostat-effect direction-fixation time of 3.01 sec. Similar models are being constructed to evaluate clinostat-based direction-fixation times for toc132-2, pgm, and lazy2,3,4 roots. Future experiments will be aimed at defining the clinostat-based direction-fixation times of the gravisensitive mutants described above, and evaluate effectiveness of this method at comparing gravisensitivities between genotypes relative to classical presentation-time methods. We will also adopt this system to evaluate sensitivity to clinostat effect (and gravisensitivity) of multiple Brachypodium distachyon accessions, with the objective of using a Genome Wide Association Study (GWAS) approach to identify loci that contribute to the variation of this trait in Brachypodium populations, hence genes that contribute to root responses to the clinostat effect and gravi-sensing in this species. Acknowledgments: This project was spearheaded by Drs. Shih-Heng Su, helped by undergraduate students Alexander D. Moen (Analysis of toc132, pgm1, arg1-3 mutant responses to clinorotation) and Tanner Zocher (Analysis of lazy and ACG20 mutant responses to clinorotation). Undergraduate students Timothy C. Davenport and Tyler Serie Anne Ganser also helped bulking up seeds from a population of Brachypodium distachyon accessions for subsequent use in GWAS analysis (next year), along with Drs. Shih-Heng Su and Patrick H. Masson. The Covid-19 lockdown interfered with progress in this project, starting mid-March 2020. We were however allowed to continue bulking up seeds from our populations of Brachypodium distachyon accessions during this period, a process that will allow us to next year initiate a GWAS analysis of Brachypodium root responses to clinorotation.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Su S-H, Moen AD, Groskopf RM, Baldwin K, Vesperman B, Masson PH. "Using Low-Speed 2-Dimensional Clinorotation to Study Gravi-sensitivity in Arabidopsis and Brachypodium." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstract Book. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Su S-H, Masson PH. "A new wrinkle in our understanding of the role played by auxin in root gravitropism." New Phytol. 2019 Oct;224(2):543-6. https://doi.org/10.1111/nph.16140 ; PMID: 31545888 , Oct-2019

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